

10.12 PERTURBATIVE QFT AS AN ASYMPTOTIC SERIES

One of the profound insights into the nature of perturbative quantum field theory (pQFT) comes not from a complex calculation, but from a simple, elegant physical argument by Freeman Dyson (1952).³ It explains *why* the power series we calculate in QFT cannot be a convergent series, but must instead be an asymptotic one.

THE PREMISE: CONVERGENCE IMPLIES ANALYTICITY

Let's consider a physical observable $\mathcal{A}$ (like a scattering amplitude) that we calculate in pQFT. We compute it as a power series in the coupling constant, g . (In QED, we often expand in $\alpha \propto e^2$).

$$\mathcal{A}(g) = \sum_{n=0}^{\infty} c_n g^n$$

c_n obtainable from the path integral on the fields. Now, let's make an assumption for the sake of contradiction: **Assume this series converges.**

If the series has a non-zero radius of convergence $R > 0$, it means the function $\mathcal{A}(g)$ is **analytic** in a disk of radius R in the complex g -plane. This is a very powerful mathematical constraint. It implies that the function $\mathcal{A}(g)$ must be well-behaved and well-defined for *all* complex values of g inside that disk. This includes small positive g , small imaginary g , and – crucially – small *negative* g .

THE PHYSICAL TEST: DYSON'S "UNSTABLE WORLD" ($g < 0$)

Dyson's genius was to ask: what would the physics of our theory look like if the coupling constant g had the opposite sign?

Let's use QED as our example, where the expansion parameter is $\alpha \propto e^2 > 0$.

• Our World ($\alpha > 0$):

- The potential between two electrons is $V(r) = +\frac{e^2}{4\pi r} > 0$. This is **repulsive**.
- The potential between an electron and a positron is $V(r) = -\frac{e^2}{4\pi r} < 0$. This is **attractive**.

³F. J. Dyson, "Divergence of perturbation theory in quantum electrodynamics," Phys. Rev. **85**, 631 (1952).

- 1765 – This configuration leads to a **stable vacuum**. The vacuum is
 1766 the state of lowest energy. Creating an e^+e^- pair costs energy
 1767 ($2m_e c^2$), and they will be attracted to each other and
 1768 annihilate.
- 1769 • **Dyson's World** ($\alpha < 0$): Now, let's see what happens if we flip the
 1770 sign of α (or equivalently, let $e^2 \rightarrow -e^2$).
- 1771 – The potential between two electrons becomes $V(r) = -\frac{e^2}{4\pi r} < 0$.
 1772 **Like charges now attract!**
- 1773 – The potential between an electron and a positron becomes
 1774 $V(r) = +\frac{e^2}{4\pi r} > 0$. **Opposite charges now repel!**

1775 THE CATASTROPHE: VACUUM INSTABILITY

1776 This new “Dyson's World” is physically pathological and catastrophically
 1777 unstable.

1778 Imagine the “empty” vacuum. A quantum fluctuation can create an e^+e^-
 1779 pair, costing $2m_e c^2$ of energy.

- 1780 1. **Creation:** An e^+e^- pair is created from the vacuum.
- 1781 2. **Repulsion:** In Dyson's World, the e^+ and e^- *repel* each other and
 1782 fly apart.
- 1783 3. **Spontaneous Boiling:** The vacuum can now create a *second* pair
 1784 (e_2^+, e_2^-). But now, e_2^- is *attracted* to e_1^- , and e_2^+ is *attracted* to e_1^+ .
- 1785 4. **Runaway Process:** The vacuum can continue to “boil,” creating a
 1786 vast number of pairs. All the electrons will clump together in one
 1787 region, and all the positrons will clump together in another region,
 1788 repelling the electron-clump and flying away.
- 1789 5. **Unbounded Negative Energy:** The potential energy of a clump of
 1790 N attracting electrons is proportional to $-N^2$. For large N it grows
 1791 faster than the rest-mass cost, proportional to N , and the kinetic
 1792 (Fermi) energy of the confined electrons, which grows like $N^{4/3}$, so
 1793 the total energy can be made arbitrarily negative.

1794 The conclusion is stark: for $g < 0$, the vacuum is not the state of lowest
 1795 energy. The Hamiltonian is unbounded below, there is no ground state,
 1796 and the would-be vacuum decays by tunneling into these pathological
 1797 states.

1798 CONNECTING PHYSICS BACK TO MATHEMATICS

1799 Here is the contradiction:

- 1800 1. Our initial assumption (that the series converges) implies $\mathcal{A}(g)$
 1801 must be **analytic** for small negative g .

1802 2. Our physical analysis (Dyson’s argument) shows that the theory is
 1803 **catastrophically unstable** for *any* small negative g . Observables
 1804 there are either undefined or acquire imaginary parts (decay rates),
 1805 so they cannot be the values of a single analytic function that is
 1806 real on both sides of $g = 0$.

1807 Therefore, our initial assumption must be false. The pQFT series **cannot**
 1808 **converge**. Dyson’s argument is heuristic, not a proof, but its conclusion
 1809 is borne out by explicit large-order calculations, starting with the toy
 1810 model below.

1811 The point $g = 0$ is not a regular point of $\mathcal{A}(g)$, so the perturbation series is
 1812 not a Taylor series with a finite radius of convergence. It is an
 1813 **asymptotic expansion**: truncated at fixed order it approximates $\mathcal{A}(g)$
 1814 better and better as $g \rightarrow 0$, but at fixed g the terms eventually grow. In
 1815 the toy model below, $g = 0$ is a **branch point**: the function is analytic in
 1816 the complex g -plane cut along the negative real axis.

1817 WHAT IS AN ESSENTIAL SINGULARITY?

1818 In complex analysis, a singularity is a point where a function is not
 1819 “well-behaved” (analytic). An essential singularity is the most
 1820 “pathological” or “wild” kind of singularity. Its behavior is so chaotic that
 1821 it doesn’t approach any single limit.
 1822 We can classify singularities at $z = 0$ using their Laurent series (a Taylor
 1823 series that allows negative powers):

- 1824 • **1. Removable Singularity (Tame):** A “patchable” hole.
 1825 – **Example:** $f(z) = \frac{\sin(z)}{z} = 1 - \frac{z^2}{3!} + \frac{z^4}{5!} - \dots$
 1826 – **Series:** Has *no* negative-power terms.
 1827 – **Behavior:** As $z \rightarrow 0$, $f(z) \rightarrow 1$. It approaches a finite limit.
- 1828 • **2. Pole (Predictable):** The function goes to infinity.
 1829 – **Example:** $f(z) = \frac{1}{z^2}$
 1830 – **Series:** Has a *finite number* of negative-power terms (here,
 1831 just z^{-2}).
 1832 – **Behavior:** As $z \rightarrow 0$, $|f(z)| \rightarrow \infty$ predictably.
- 1833 • **3. Essential Singularity (Wild):** The function is chaotic.
 1834 – **Example:** $f(z) = e^{1/z}$
 1835 – **Series:** $f(z) = 1 + \frac{1}{z} + \frac{1}{2!z^2} + \frac{1}{3!z^3} + \dots$
 1836 – **Definition:** Has an *infinite number* of negative-power terms.
 1837 – **Behavior:** The limit as $z \rightarrow 0$ does not exist. It depends on
 1838 the path:
 1839 * Path 1 (Real axis, $z = x \rightarrow 0^+$): $e^{1/x} \rightarrow \infty$.
 1840 * Path 2 (Real axis, $z = x \rightarrow 0^-$): $e^{1/x} \rightarrow e^{-\infty} \rightarrow 0$.

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* Path 3 (Imaginary axis, $z = iy \rightarrow 0$):
 $e^{1/(iy)} = e^{-i/y} = \cos(1/y) - i \sin(1/y)$, which oscillates
infinitely fast and approaches no limit.

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Connection to QFT: Non-perturbative effects often have the form
 $f(g) = e^{-c/g^2}$ (or $e^{-c/g}$ when g is the square of a coupling). This function
has an essential singularity at $g = 0$. If you try to calculate its Taylor
series, you find that the function and all its derivatives are exactly zero at
 $g = 0$. Its Taylor series is just $0 + 0g + 0g^2 + \dots$, which clearly does not equal
the function. This is why such terms are “invisible” to perturbation
theory.

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A TOY MODEL: 0-DIMENSIONAL ϕ^4 THEORY

We can see this mathematical behavior in a simple integral, which we can
think of as a 0-dimensional QFT. The “path integral” for a $g\phi^4$ theory is:

$$Z(g) = \int_{-\infty}^{\infty} d\phi \exp\left(-\frac{1}{2}\phi^2 - \frac{g}{4!}\phi^4\right)$$

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This $Z(g)$ represents the “vacuum” of the theory.

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- **Our World ($g > 0$):** The $e^{-g\phi^4}$ term helps the $e^{-\phi^2}$ term, making the
integral *even more* convergent for large ϕ . The theory is stable.
- **Dyson’s World ($g < 0$):** Let $g = -|\lambda|$ where $|\lambda| > 0$.

$$Z(-|\lambda|) = \int_{-\infty}^{\infty} d\phi \exp\left(-\frac{1}{2}\phi^2 + \frac{|\lambda|}{4!}\phi^4\right)$$

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As $\phi \rightarrow \pm\infty$, the $+\phi^4$ term in the exponent dominates. The $e^{+\phi^4}$ term
explodes, and the integral **diverges** completely.

The integral defines $Z(g)$ for $g > 0$ and diverges for all real $g < 0$. In fact it
can be done exactly,

$$Z(g) = \sqrt{\frac{3}{g}} e^{3/(4g)} K_{1/4}\left(\frac{3}{4g}\right),$$

with $K_{1/4}$ a modified Bessel function. Rotating the integration contour
continues $Z(g)$ into the complex plane, but the continuations to negative
 g from above and from below do not agree: they differ by an imaginary
part proportional to $e^{-3/(2|g|)}$. So the negative real axis is a branch cut and
 $g = 0$ is a branch point. Expanding the exponential in g gives

$$Z(g) \sim \sum_n c_n g^n, \quad c_n = \sqrt{2\pi} \frac{(-1)^n (4n-1)!!}{24^n n!}, \quad \frac{c_{n+1}}{c_n} \rightarrow -\frac{2}{3}n,$$

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so $c_n \sim (-2/3)^n n!$ up to powers of n . Factorial growth is the signature of
an asymptotic series with zero radius of convergence.

FINDING THE $e^{-c/g}$ TERM (AN INSTANTON CALCULATION)

How do we know the non-analytic part has the form $e^{-c/g}$? This term doesn't just appear; it emerges from the path integral when we look beyond the "trivial" solution. We can find it using a **saddle-point approximation** (the 0-dimensional version of an "instanton" calculation). The integral is $Z(g) = \int d\phi e^{-S(\phi)}$, where the "action" is $S(\phi) = \frac{1}{2}\phi^2 + \frac{g}{24}\phi^4$. The dominant contributions to this integral come from points where the action is stationary, i.e., $S'(\phi) = 0$.

$$S'(\phi) = \frac{dS}{d\phi} = \phi + \frac{g}{6}\phi^3 = \phi \left(1 + \frac{g}{6}\phi^2\right) = 0$$

This equation has several solutions (saddle points):

- **The Trivial Saddle:** $\phi_0 = 0$. If we approximate the integral by expanding $S(\phi)$ around $\phi = 0$, we are just doing standard perturbation theory. This gives the asymptotic series $\sum c_n g^n$.
- **The Non-Trivial Saddles:** $1 + \frac{g}{6}\phi^2 = 0$. This gives $\phi_{inst}^2 = -6/g$. These are the "instanton" solutions.
 - For $g > 0$ (our world), ϕ_{inst}^2 is negative, so these solutions $\phi_{\pm} = \pm i\sqrt{6/g}$ are in the complex plane. They aren't on our real integration path.
 - For $g < 0$ (Dyson's world), let $g = -|\lambda|$. Then $\phi_{inst}^2 = -6/(-|\lambda|) = 6/|\lambda|$. The solutions $\phi_{\pm} = \pm\sqrt{6/|\lambda|}$ are **real**. They are maxima of S ($S'' = -2$ there): the tops of the barrier that the unstable vacuum must tunnel through. Their action, $3/(2|\lambda|)$, sets the decay rate below.

The contribution of any saddle point to the integral is proportional to $e^{-S_{saddle}}$. Let's calculate the action at these new saddles:

$$S(\phi_{inst}) = \frac{1}{2}\phi_{inst}^2 + \frac{g}{24}\phi_{inst}^4$$

Substitute $\phi_{inst}^2 = -6/g$:

$$S(\phi_{inst}) = \frac{1}{2} \left(-\frac{6}{g}\right) + \frac{g}{24} \left(-\frac{6}{g}\right)^2$$

$$S(\phi_{inst}) = -\frac{3}{g} + \frac{g}{24} \left(\frac{36}{g^2}\right) = -\frac{3}{g} + \frac{3}{2g}$$

$$S(\phi_{inst}) = -\frac{3}{2g}$$

Therefore, the contribution to the integral from these saddles is:

$$Z_{non-pert} \sim e^{-S(\phi_{inst})} = e^{-(-3/(2g))} = e^{3/(2g)}$$

1882 This is the non-analytic behavior. What it does depends on the sign of g :

- **For** $g = -|\lambda|$, as in Dyson's unstable world, the saddles are real and their contribution is

$$Z_{non-pert} \sim e^{3/(2(-|\lambda|))} = e^{-3/(2|\lambda|)}.$$

1883 This is the $e^{-c/|g|}$ form, and it is exactly the size of the discontinuity
1884 across the cut found above: the imaginary part of Z on the
1885 unstable side, the analogue of a vacuum decay rate.

- **For** $g > 0$ the saddles $\pm i\sqrt{6/g}$ are complex and the integration
1886 contour does not pass through them, so they do not contribute to
1887 $Z(g)$ (the factor $e^{+3/(2g)}$ would be enormous). They are still felt: the
1888 same saddle action fixes how fast the perturbative coefficients grow.
1889 The large-order rate $c_{n+1}/c_n \rightarrow -\frac{2}{3}n$ is n divided by $-\frac{3}{2}$, the saddle
1890 action $S = -3/(2g)$ at $g = 1$.
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1892 Either way, $e^{-3/(2|g|)}$ is non-perturbative: all its derivatives vanish as
1893 $g \rightarrow 0^-$, so it is completely "invisible" to the expansion around $g = 0$.